SCORING SF-12v2™ SCALES AND SUMMARY MEASURES

This document provides "**pre-released**" scoring instructions for the eight scales and two summary measures in the SF-12 Health Survey, Version 2.0. General scoring information and steps for data entry and scoring that are common to all items are discussed first (see Figure 1). Next, formulas for item aggregation and transformation of scale scores on a 0-100 scale are presented. Following the 0-100 transformation is the norm-based scoring of each scale to have a mean of 50 and a standard deviation of 10 in the 1998 general U.S. population. Finally, algorithms for computing the physical and mental health summary scales are provided.

The scoring algorithms presented in this document are considered "pre-release" since evaluations of various missing data estimation methods and scoring algorithms are still ongoing. There may be slight modifications in the "pre-released" scoring algorithms after deciding upon the most appropriate missing data estimation method and scoring algorithm. The slight modifications of the final scoring algorithms will not change conclusions of study results based on the "pre-released" version of the scoring algorithms, since in the many datasets used to evaluate the various missing data estimation methods and scoring algorithms, the correlation between scores estimated with the various methods exceeded 0.97 for each scale. Any difference will be observed in the absolute score value of the scale.

Importance of Standardization

As with all standardized tests, standardization of content and scoring is what makes interpretation of SF-12® scales and summary measures possible. The content of the SF-12 form and the "pre-released" scoring algorithms were selected and standardized following careful study of many options. The algorithms described in this document were chosen to be as simple as possible while still satisfying the assumptions of the methods used to construct SF-12 scales and summary measures.

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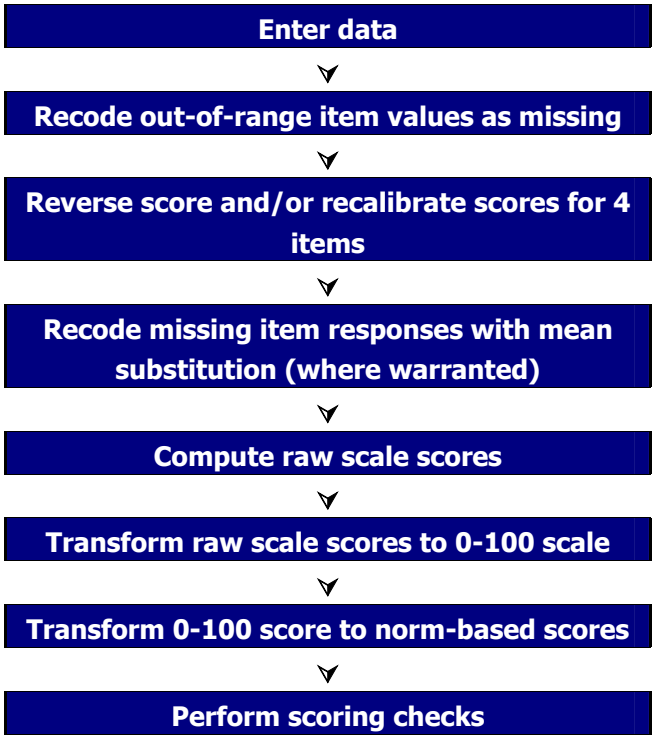
Pre-release scoring algorithms (N = 6,859 to 7,018)

Figure 1. **Flow Chart for Scoring SF-12v2™ Scales**



General Scoring Information

SF-12v2 items and scales are scored so that a higher score indicates a better health state. For example, functioning scales are scored so that a high score indicates better functioning and the pain scale is scored so that a high score indicates freedom from pain. After data entry, items and scales are scored in four steps:

1. Item recoding for the 4 items that require recoding
2. Computing scale scores by summing across items in the same scale
   (raw scale scores)
3. Transforming raw scale scores to a 0 - 100 scale (transformed scale scores)
4. Transforming 0-100 scale scores to have a mean of 50 and standard deviation of 10 in the general U.S. population (norm-based scale scores).

Tables 1 through 8 present scoring information for the items used in each of the eight SF-12 health scales. Each table presents the verbatim content of each question, response choices, and both the precoded values printed in the questionnaire and final values for scoring each item. Item numbers in Tables 1 through 8 correspond to those on the Standard SF-12v2 form.

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Table 1 Physical Functioning: Verbatim Items and Scoring Information

Verbatim Items

2. Moderate activities, such as moving a table, pushing a vacuum cleaner, bowling, or playing golf
3 Climbing several flights of stairs

Precoded and Final Values for Items 2 and 3

Response Choices	*Precoded Item Value*	*Final Item Value*
Yes, limited a lot	1	1
Yes, limited a little	2	2
No, not limited at all	3	3

Scale Scoring

Compute the simple algebraic sum of the final item scores as shown in Table 10. See text for handling of missing item responses. This scale is scored so that a high score indicates better physical functioning.

Note: Precoded values are as shown on the appended form.
This scale does not require recoding of items prior to computation of the scale score.

Table 2 Role-Physical: Verbatim Items and Scoring Information

Verbatim Items

4. **Accomplished less** than you would like
5. Were limited in the **kind** of work or other activities

Precoded and Final Values for Items 4 - 5

Response Choices	*Precoded Item Value*	*Final Item Value*
All of the time	1	1
Most of the time	2	2
Some of the time	3	3
A little of the time	4	4
None of the time	5	5

Scale Scoring

Compute the simple algebraic sum of the final item scores as shown in Table 10. See text for handling of missing item responses. This scale is scored so that a high score indicates better Role-Physical functioning.

Note: Precoded values are as shown on the appended form.
This scale does not require recoding of items prior to computation of the scale score.

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Table 3 Bodily Pain: Verbatim Items and Scoring Information

Verbatim Items

8. During the past 4 weeks, how much did pain interfere with your normal work
   (including both work outside the home and housework)?

Precoded and Final Values for Item 8

Response Choices	*Precoded Item Value*	*Final Item Value*
Not at all	1	5
A little bit	2	4
Moderately	3	3
Quite a bit	4	2
Extremely	5	1

Scale Scoring

Instructions for scoring the BP scale are presented in Table 10. This scale is scored so that a high score indicates lack of bodily pain.

Note: Precoded values are as shown on the appended form.
This scale requires recoding of the item response values prior to computation of the scale score.

Table 4 General Health: Verbatim Items and Scoring Information

Verbatim Items

1. In general, would you say your health is:

Precoded and Final Values for Item 1

Response Choices	*Precoded Item Value*	*Final Item Value*
Excellent	1	5.0
Very good	2	4.4
Good	3	3.4
Fair	4	2.0
Poor	5	1.0

Scale Scoring

Instructions for scoring the GH scale are presented in Table 10. This scale is scored so that a high score indicates better general health perceptions.

Note:Precoded values are as shown on the appended form.
This scale requires recoding of the item response values prior to computation of the scale score.

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Table 5 Vitality: Verbatim Items and Scoring Information

Verbatim Items

10. Did you have a lot of energy?

Precoded and Final Values for Item 10

Response Choices	*Precoded Item Value*	*Final Item Value*
All of the time	1	5
Most of the time	2	4
Some of the time	3	3
A little of the time	4	2
None of the time	5	1

Scale Scoring

Instructions for scoring the VT scale are presented in Table 10. This scale is scored so that a high score indicates more vitality.

Note: Precoded values are as shown on the appended form.
This scale requires recoding of the item response values prior to computation of the scale score.

Table 6 Social Functioning: Verbatim Items and Scoring Information

Verbatim Items

12. During the **past 4 weeks**, how much of the time has your physical health or emotional problems interfered with your social activities (like visiting with friends, relatives, etc.)?

Precoded and Final Values for Items 12

Response Choices	*Precoded Item Value*	*Final Item Value*
All of the time	1	1
Most of the time	2	2
Some of the time	3	3
A little of the time	4	4
None of the time	5	5

Scale Scoring

Instructions for scoring the SF scale are presented in Table 10. This scale is scored so that a high score better social functioning.

Note: Precoded values are as shown on the appended form.
This scale does not require recoding of the item response values prior to computation of the scale score.

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Table 7 Role-Emotional: Verbatim Items and Scoring Information

Verbatim Items

6. **Accomplished less** than you would like
7. Did work or other activities less **carefully** than usual

Precoded and Final Values for Items 6 & 7

Response Choices	*Precoded Item Value*	*Final Item Value*
All of the time	1	1
Most of the time	2	2
Some of the time	3	3
A little of the time	4	4
None of the time	5	5

Scale Scoring

Compute the simple algebraic sum of the final item scores as shown in Table 10. See text for handling of missing item responses. This scale is scored so that a high score indicates better Role-Emotional functioning.

Note: Precoded values are as shown on the appended form.
This scale does not require recoding of items prior to computation of the scale score.

Table 8 Mental Health: Verbatim Items and Scoring Information

Verbatim Items

9. Have you felt calm and peaceful?
11. Have you felt downhearted and depressed?

Precoded and Final Values for Items 9 & 11

Item 9	*Response Choices*	*Precoded Item Value*	*Final Item Value*
	All of the time	1	5
	Most of the time	2	4
	Some of the time	3	3
	A little of the time	4	2
	None of the time	5	1

Item 11	*Response Choices*	*Precoded Item Value*	*Final Item Value*
	All of the time	1	1
	Most of the time	2	2
	Some of the time	3	3
	A little of the time	4	4
	None of the time	5	5

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Scale Scoring

Compute the simple algebraic sum of the final item scores as shown in Table 10. See text for handling of missing item responses. This scale is scored so that a high score indicates better mental health.

Note: Precoded values are as shown on the appended form.
This scale requires recoding of one item prior to computation of the scale score.

Item Recoding

The next stage after data entry is the recoding of response choices as shown in Tables 1 through 8. Item recoding is the process of deriving the item values that will be used to calculate the scale scores. Several steps are included in this process: (1) change out-of-range values to missing, (2) recode values for 10 items, and (3) substitute person-specific estimates for missing items.

Out-of-Range Values

All 12 items should be checked for out-of-range values prior to assigning the final item values. Out-of-range values are those that are lower than an item's precoded minimum value, or higher than an item's precoded maximum value (see Tables 1 through 8). Out-of-range values are usually caused by data-entry errors and, if possible, should be changed to the correct response through verification with the original questionnaire. If the questionnaire is not available, all out-of-range values should be recoded as missing data.

Recode Values for Ten Items

Four items are reverse scored. Reverse scoring of items is done to ensure that a higher item value indicates better health on all SF-12v2 items. SF-12v2 items that need to be reverse scored are worded so that a higher precoded item value indicates a poorer health state.

Item Recalibration

For 11 of the SF-12v2 items, research to date offers good support for the assumption of a linear relationship between item scores and the underlying health concept defined by their scales. However, empirical work has shown that one item (General Health, question 1) requires recalibration to satisfy this important scaling assumption.

General Health Rating Item

The "Very Good" and "Good" responses to Item 1 (GH1) are recalibrated to achieve a better linear fit with the general health evaluation concept measured by the GH scale. Empirical studies during the Health Insurance Experiment (HIE) were among the first to document that the intervals between response choices for this item are not equal (Davies & Ware, 1981). Subsequent studies of Item 1 (GH1), using both the Thurstone Method of Equal-Appearing Intervals (Thurstone & Chave, 1929) and other empirical methods, have also consistently shown that the interval between "Excellent" and "Very Good" is about half the size of the interval between "Fair" and "Good" (Ware, Nelson et al., 1992). These results have been confirmed in studies of SF-36® translations from 10 countries

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participating in the International Quality of Life Assessment (IQOLA) project. Finally, in all studies we are aware of to date, mean values for a criterion general health scale for respondents who choose each of the five levels defined by Item 1 (GH1) depart significantly from linearity.

Results from the Medical Outcome Study (MOS) that served as the basis for the recommended recalibration of Item 1 (GH1) are summarized in Table 9. As shown in Table 9 and discussed elsewhere (Ware, Nelson, et al., 1992), the mean criterion scores were remarkably similar for those who chose the same category of Item 1 across the screening (N=18,573) and longitudinal (N=3,054) samples. Intervals between adjacent response categories were unequal, as observed in the HIE (Davies & Ware, 1981). For these reasons, item scale values are transformed as shown in Table 9 using specific results from the screening sample. The result is a very high 0.70 correlation with the sum of the other four items in the GH scale.

Table 9 **Mean Current Health Scores for Respondents Choosing Each Level of SF-36 Item 1 (GH1)**

	Mean Current Health		Recommended Scoring	
Response to Item 1	Screening Sample (N=18573)	Baseline Sample (N=3054)	1-5 Scale	0-100 Scale
Excellent	87.9	86.9	5.0	100
Very good	75.5	75.4	4.4	84
Good	57.6	55.9	3.4	61
Fair	30.0	30.6	2.0	25
Poor	10.8	10.8	1.0	0

Note: Adapted from: Ware, Nelson et al. (1992). Preliminary tests of a 6-item general health survey: A patient application. In A.L. Stewart & J.E. Ware (Eds.), Measuring functioning and well-being: The Medical Outcomes Study approach (p. 299). Durham, NC: Duke University Press.

How to Treat Missing Data

Various options for handling missing data are currently being evaluated for the four SF-12v2 scales with two items each (PF, RP, RE and MH). Until our evaluation of the various options is complete, we recommend substituting the response value of the completed item for the item with a missing response in each two-item scale. Scale scores are set to missing when there are missing response values for the four single item scales (BP, GH, VT, and SF).

Computing Raw Scale Scores

After item recoding, including handling of missing data, a raw score is computed for each scale. This score is the simple algebraic sum of

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responses for all items in that scale, as shown in Table 10. For example, the raw scale score for the Role-Physical Scale is the sum of the scores for Items 4 and 5. For single item scales, the raw score is simply the final item response value. Use recoded item values and imputed values where applicable.

This simple scoring method is possible because items in the same scale have roughly equivalent relationships to the underlying health concept being measured and no item is used in more than one scale. Thus, it is not necessary to standardize or weight items. These assumptions have been extensively tested and verified (McHorney et al., 1994).

Transformation of Scale Scores

The next step involves transforming each raw scale score to a 0 - 100 scale using the formula shown below. Table 10 provides the information necessary to apply this formula to each scale.

$$\text{Transformed Scale} = \left[\frac{\text{(Actual raw score - lowest possible raw score)}}{\text{Possible raw score range}}\right] * 100$$

This transformation converts the lowest and highest possible scores to zero and 100, respectively. Scores between these values represent the percentage of the total possible score achieved.

Table 10 **Formulas for Scoring and Transforming Scales**

SF-12v2 Scale	Sum Final Item Values (after recoding items as in Table 1-8)	Lowest and highest possible raw scores	Possible raw score range
Physical Functioning	2+3	2, 6	4
Role-Physical	4+5	2, 10	8
Bodily Pain	8	1, 5	4
General Health	1	1, 5	4
Vitality	10	1, 5	4
Social Functioning	12	1, 5	4
Role-Emotional	6+7	2, 10	8
Mental Health	9+11	2, 10	8

Formula and example for transformation of raw scale scores to 0-100 scale scores

⌈ (Actual raw score - lowest *possible* raw score) ⌉

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$$\text{Transformed Scale} = \frac{\quad}{\lfloor \textit{Possible}\ \text{raw score range} \rfloor} * 100$$

Example: A Physical Functioning raw score of 5 would be transformed as follows:

$$\frac{\lceil (5 - 2) \rceil}{\lfloor 4 \rfloor} * 100 = 75$$

Where lowest possible score = 2 and possible raw score range = 4

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Norm-Based Scoring of Scale Scores

The next step involves the norm-based scoring of each 0 - 100 scale score using the formulas shown below. Table 11 provides the information necessary to apply these formulas to each SF-12v2 scale. The means and standard deviations (Table 11) used in norm-based scoring come from the 1998 general U.S. population. A linear z-score transformation is used so that all eight SF-12v2 scales have a mean of 50 and a standard deviation of 10 in the 1998 general U.S. population.

The advantage of the standardization and norm-based scoring of the 8 SF-12v2 scales is that results for one scale can be meaningfully compared with the other scales and their scores have a direct interpretation in relation to the distribution of scores in the 1998 general U.S. population. Specifically, all scores above or below 50 are above or below the average, respectively, in the 1998 general U.S. population. Because the standard deviation is 10 for all 8 scales, each one point difference or change in scores also has a direct interpretation. A one point difference or change is one-tenth of a standard deviation unit or an effect size of 0.10. Lastly, norm-based scoring provides the basis for comparing scale scores across Version 1.0 and Version 2.0 standard forms.

The first step in norm-based scoring consists of standardizing each SF-12v2 scale using a z-score transformation. A z-score for each scale is computed by subtracting the 1998 general U.S. population mean (Table 11) for each SF-12v2 scale and dividing the difference by the corresponding scale standard deviation (Table 11) from the 1998 general U.S. population. Formulas are listed below.

Step 1. Formulas for z-score standardization of SF-12v2 scales

PF_Z = (PF – 81.25221) / 29.25167
RP_Z = (RP – 80.55383) / 27.2212
BP_Z = (BP – 81.74015) / 24.53019
GH_Z = (GH – 72.19795) / 23.19041
VT_Z = (VT – 55.59090) / 24.84380
SF_Z = (SF – 83.73973) / 24.75775
RE_Z = (RE - 86.43625) / 22.48285
MH_Z = (MH – 70.21132) / 20.64230

Means and standard deviations are from Table 11.

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Table 11 **1998 General U.S. Population Means and Standard Deviations Used to Derive SF-12v2 z-scores**

SF-12v2 Scale	Mean	SD
PF	81.25221	29.25167
RP	80.55383	27.22120
BP	81.74015	24.53019
GH	72.19795	23.19041
VT	55.59090	24.84380
SF	83.73973	24.75775
RE	86.43625	22.48285
MH	70.21132	20.64230

Norm-based Transformation of SF-12v2Z-Scores

The second step involves transforming each SF-12v2 z-score to the norm-based (50, 10) scoring. This is accomplished by multiplying each z-score by 10 and adding the resulting product to 50. Formulas are listed below.

Step 2. Norm-based transformation of SF-12v2 z-scores:

Norm-Based PF: PF = 50 + (PF_Z * 10)

Norm-Based RP: RP = 50 + (RP_Z * 10)

Norm-Based BP: BP = 50 + (BP_Z * 10)

Norm-Based GH:GH = 50 + (GH_Z * 10)

Norm-Based VT: VT = 50 + (VT_Z * 10)

Norm-Based SF: SF = 50 + (SF_Z * 10)

Norm-Based RE: RE = 50 + (RE_Z * 10)

Norm-Based MH: MH = 50 + (MH_Z * 10)

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SCORING SF-12v2 PHYSICAL AND MENTAL SUMMARY MEASURES

Scoring of the SF-12v2 PCS and MCS summary measures involves three steps. First, the eight SF-12v2 scales are standardized using means and standard deviations from the 1998 general U.S. population. Second, they are aggregated using weights (factor score coefficients) from the 1990 general U.S. population. These are the same weights as those used to score PCS and MCS from the SF-36 Version 1.0 (Ware et al., 1994). Finally aggregate PCS and MCS scores are standardized using a linear T-score transformation to have a mean of 50 and a standard deviation of 10, in the 1998 general U.S. population.

General U.S. population statistics used in the standardization and in the aggregation of SF-12v2 scale scores are presented in Table 11 above and repeated in Table 12 below. Detailed information including formulas for scale aggregation and transformation of scores are presented below. Formal checks using a test dataset made available upon request at QualityMetric can be performed to confirm the successful reproduction of PCS and MCS scales, as discussed later in the chapter. We strongly recommend these tests.

Importance of Standardization

As with the scoring of SF-12v2 items and scales, which are aggregated to score the summary measures, standardization of the scoring of the PCS and MCS scales is vital to their interpretation. Any changes in scoring of the SF-12v2 items, scales, or the algorithms for the summary scales may compromise their reliability and validity. Changes in scoring have also been shown to invalidate normative comparisons, and changes are likely to complicate or prevent meaningful comparisons of results across studies.

Norm-Based Scoring of PCS and MCS

Norm-Based Scoring

The SF-12v2 PCS and MCS scales are scored using norm-based methods. The means and standard deviations used in scoring come from the 1998 general U.S. population and the factor score coefficients come from the 1990 general U.S. population (Ware et al., 1994). A linear T-score transformation method is used so that both the PCS and MCS have a mean of 50 and a standard deviation of 10 in the 1998 general U.S. population.

The advantage of the standardization and norm-based scoring of the

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PCS and MCS is that results for one can be meaningfully compared with the other and their scores have a direct interpretation in relation to the distribution of scores in the general U.S. population. Specifically, all scores above and below 50 are above and below the average, respectively, in the 1998 general U.S. population. Because the standard deviation is 10 for both PCS and MCS measures, each one point difference in scores also has a direct interpretation. A one point difference is one-tenth of a standard deviation.

Table 12 **1998 General U.S. Population Means, Standard Deviations and 1990 Factor Score Coefficients used to Derive PCS and MCS Scale Scores, Standard Form**

SF-12v2 Scale	Mean*	SD*	Factor Score Coefficients	
			PCS	MCS
PF	81.25221	29.25167	0.42402	-0.22999
RP	80.55383	27.22120	0.35119	-0.12329
BP	81.74015	24.53019	0.31754	-0.09731
GH	72.19795	23.19041	0.24954	-0.01571
VT	55.59090	24.84380	0.02877	0.23534
SF	83.73973	24.75775	-0.00753	0.26876
RE	86.43625	22.48285	-0.19206	0.43407
MH	70.21132	20.64230	-0.22069	0.48581

**Note: The means and standard deviations for each SF-12v2 are based on the 0-100 scoring.*

Steps in Scoring

Following the scoring of the eight scales according to the standard SF-36v2™ scoring algorithms (0-100 scale) explained earlier, PCS and MCS are scored in three steps as explained below:

Standardization of Scales (Z-Scores), Standard Form

First, each SF-12v2 scale is standardized using a z-score transformation using SF-12v2 scale means and standard deviations from the 1998 general U.S. population as given in Table 12. A z-score for each scale is computed by subtracting the 1998 general U.S. population mean from each SF-12v2 scale score (0-100 scale) and dividing the difference by the corresponding scale standard deviation (0-100 scale) from the 1998 general U.S. population. Note that the SF-12v2 scales scored on the 0-100 scale are used in step 1. Norm-based SF-12v2 scale scores are not used in this step. Formulas are listed below.

Step 1. Formulas for z-score standardization of SF-36v2 scales, (Standard Form)

PF_Z = (PF - 81.25221) / 29.25167

RP_Z = (RP - 80.55383) / 27.22120

BP_Z = (BP – 81.74015) / 24.53019

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GH_Z = (GH – 72.19795) / 23.19041
VT_Z = (VT – 55.59090) / 24.84380
SF_Z = (SF - 83.73973) / 24.75775
RE_Z = (RE - 86.43625) / 22.48285
MH_Z = (MH – 70.21132) / 20.64230

Means and standard deviations are from Table 12.

Aggregation of Scale Scores (Standard Form)

After a z-score has been computed for each SF-12v2 scale, the second step involves computation of aggregate scores for the physical and mental components using the physical and mental factor score coefficients from the 1990 general U.S. population as given in Table 12.

Computation of an aggregate physical component score consists of multiplying each SF-12v2 scale z-score by its respective physical factor score coefficient and summing the eight products, as shown below. Similarly, an aggregate mental component score is obtained by multiplying each SF-12v2 scale z-score by its respective mental factor score coefficient and summing the eight products.

Step 2. Formulas for aggregating scales in estimating aggregate physical and mental component scores

AGG_PHYS = (PF_Z * .42402) + (RP_Z * .35119) + (BP_Z * .31754) + (GH_Z * .24954) + (VT_Z * .02877) + (SF_Z * -.00753) + (RE_Z * -.19206) + (MH_Z * -.22069)

AGG_MENT = (PF_Z * -.22999) + (RP_Z * -.12329) + (BP_Z * -.09731) + (GH_Z * -.01571) + (VT_Z * .23534) + (SF_Z * .26876) + (RE_Z * .43407) + (MH_Z * .48581)

Transformation of Summary Scores

The third step involves transforming each component score to the norm-based (50, 10) scoring. This is accomplished by multiplying each aggregate component scale score by 10 and adding the resulting product to 50. Formulas are listed below.

Step 3. Formulas for T-score transformation of component scores

Transformed Physical (PCS) = 50 + (AGG_PHYS * 10)
Transformed Mental (MCS) = 50 + (AGG_MENT * 10)

Missing Data Estimation

Results from ongoing evaluations of options to score PCS and MCS when a respondent is missing any one of the eight SF-12v2 scales has shown considerable promise. Evaluation of missing data rates across general and clinical populations has shown that 50% of those who have missing

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PCS and MCS scores are missing PCS and MCS because of missing data on one SF-12 scale. QualityMetric's Missing Data Estimator (MDE) enables you to score PCS and MCS with data for 7 of the 8 SF-12 scales. PCS and MCS scores from MDE have proven to be reliable and valid (Ware et al., forthcoming). Further efforts are underway to evaluate options for scoring PCS and MCS when more than 1 SF-12 scale score is missing.

Until QualityMetric's MDE is made available, it is recommended that component scale scores be set to missing if the respondent is missing any one of the eight SF-12v2 scales. To minimize the number of component scores missing, we recommend estimating each of the eight scale scores if half or more of the items are complete, as documented above.

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